

Design Calculations Report

CIV102 Bridge Project

1 Hand Calculations for Design 0 under 400N Load Case 1

Figures 1, 2, and 3 show the hand calculations for Design 0.

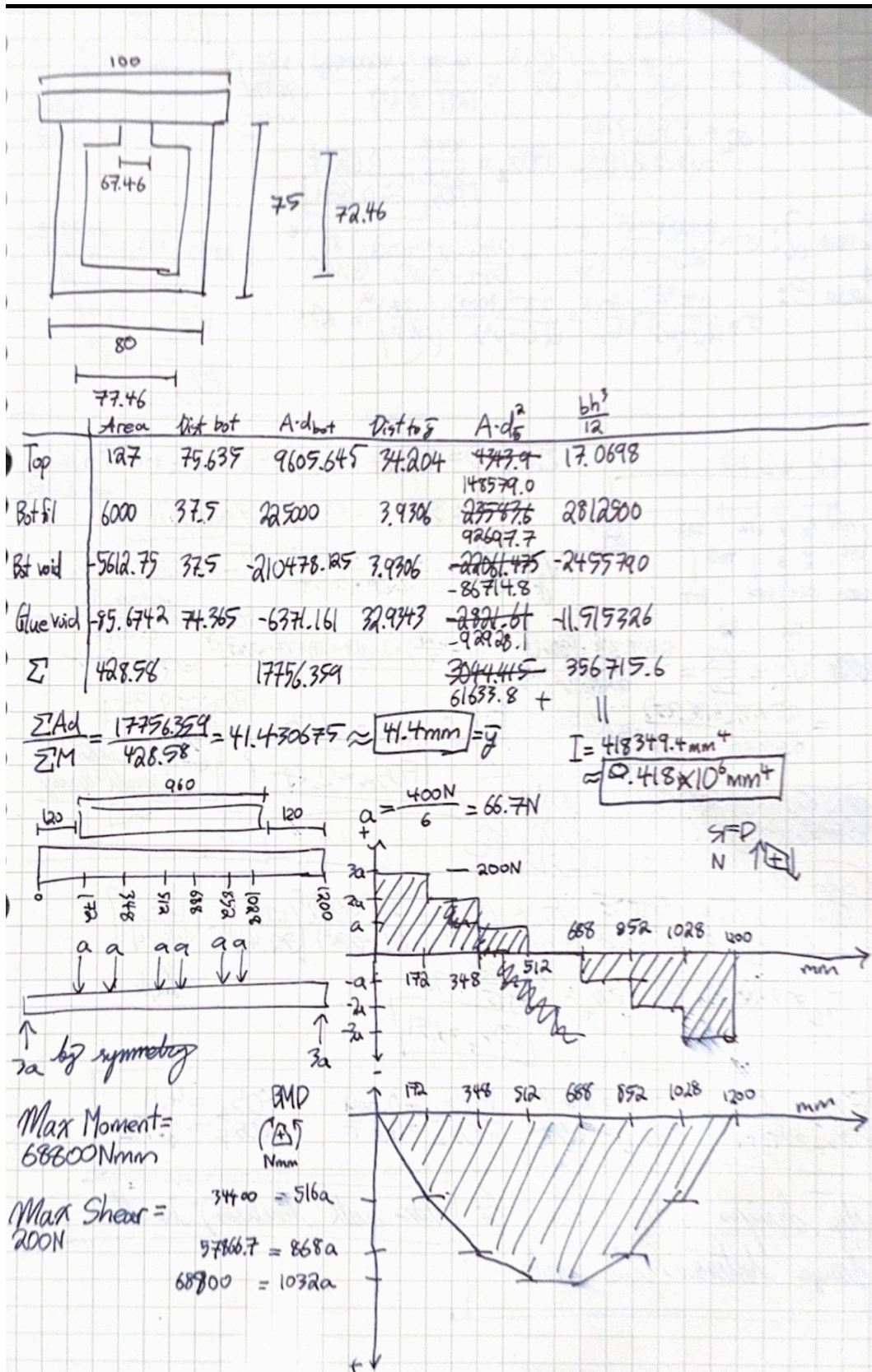


Figure 1: First page of hand calculations

Max Moment: 68800 Nmm Max Shear: 200 N $I = 0.418 \times 10^6 \text{ mm}^4$

$\sigma_{xc} = \frac{M_y}{I} = \frac{68800 \text{ Nmm} \cdot 34.87 \text{ mm}}{0.418 \times 10^6 \text{ mm}^4} = 5.73936 \approx 5.74 \text{ MPa}$

$\sigma_x = \frac{M_y}{I} = \frac{68800 \text{ Nmm} \cdot 41.4 \text{ mm}}{0.418 \times 10^6 \text{ mm}^4} = 6.81416 \approx 6.81 \text{ MPa}$

$\tau_g = \frac{VQ}{Ib} = \frac{257.71 \text{ N} \cdot 127 \text{ mm}^2 \cdot 34.235 \text{ mm}}{0.418 \times 10^6 \text{ mm}^4 \cdot 12.54 \text{ mm}} = 0.21344 \text{ MPa}$

$\tau_m = \frac{VQ}{Ib} = \frac{257.71 \text{ N} \cdot [80 \cdot 1.27 \text{ mm}^2 \cdot (41.4 \text{ mm} - 1.27/2 \text{ mm}) + 2 \cdot (41.4 - 1.27) \cdot 1.27 \text{ mm}^2]}{0.418 \times 10^6 \text{ mm}^4 \cdot 1.27 \text{ mm} \cdot 2} = 10.7791 \text{ MPa}$

$\text{FOS}_c = \frac{6}{5.74} = 1.04524$
 $\text{FOS}_c \approx 1.045$

$\text{FOS}_g = \frac{30}{6.81} = 4.40528$
 $\text{FOS}_g \approx 4.41$

$\text{FOS}_m = \frac{4}{0.21344} = 18.741$
 $\text{FOS}_m = 18.741$

$\text{FOS}_g = \frac{2}{0.21344} = 9.3699$
 $\text{FOS}_g \approx 9.37$

$\text{FOS}_H = \frac{4}{1.4995} = 2.6675$
 $\text{FOS}_H = 2.67$

$\text{FOS}_H = 1.4995$

Figure 2: Second page of hand calculations

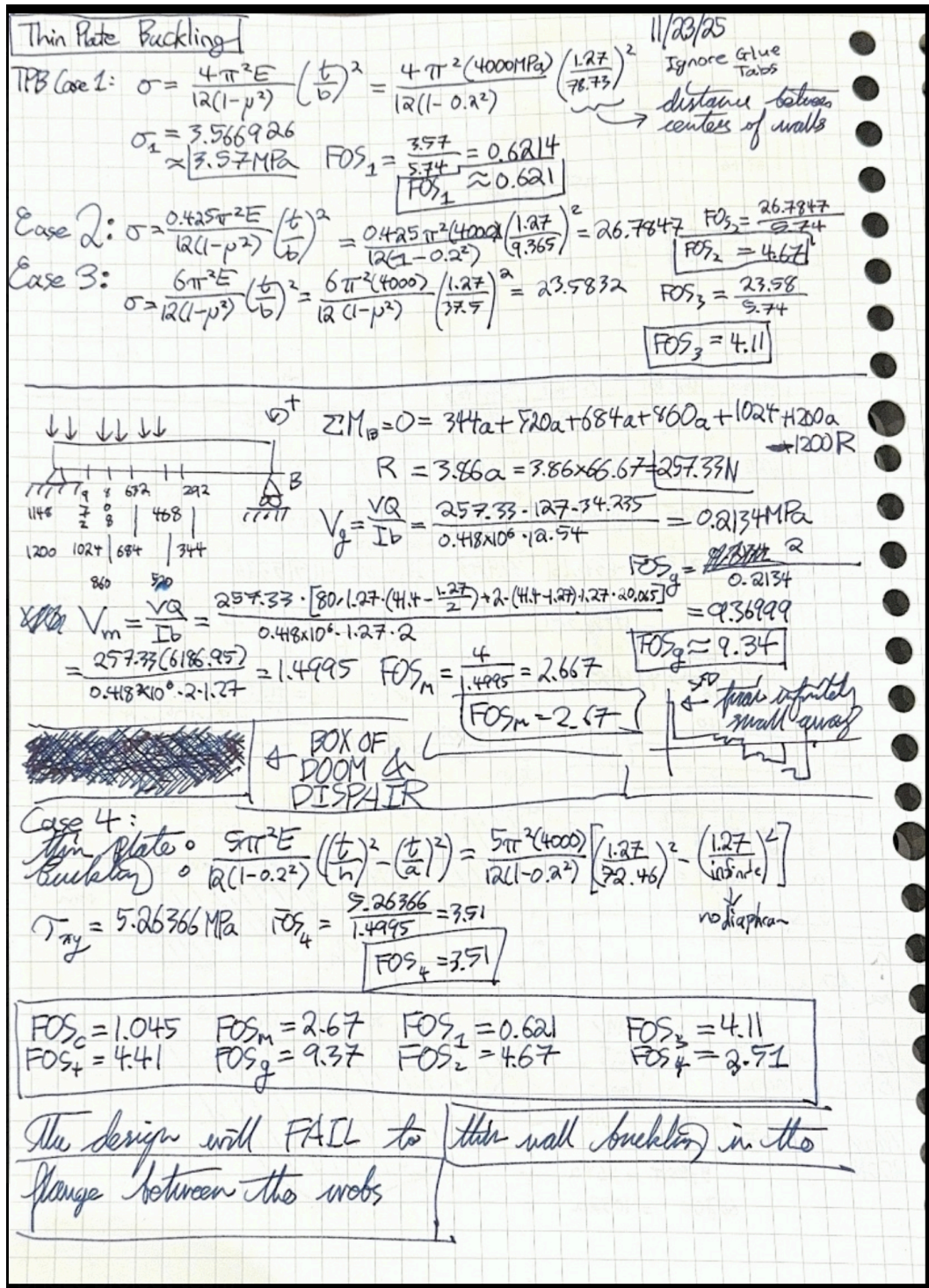


Figure 3: Third page of hand calculations

2 Evidence of Programming

2.1 Code Output for Design 0 under 400N Load Case 1

Section Properties

$$A = 428.5742 \text{ mm}^2$$

$$\bar{y} = 41.431 \text{ mm from bottom}$$

$$I = 418352.21 \text{ mm}^4$$

Support Reactions (first wheel at 197mm)

$$R_A \text{ (at 25mm)} = 200.0 \text{ N}$$

$$R_B \text{ (at 1225mm)} = 200.0 \text{ N}$$

Shear Force Envelope (SFE)

$$V_{\max} = 257.27 \text{ N at } x = 25.00 \text{ mm}$$

$$V_{\min} = -257.27 \text{ N at } x = 1224.87 \text{ mm}$$

Bending Moment Envelope (BME)

$$M_{\max} = 69443.92 \text{ N} \cdot \text{mm at } x = 582.31 \text{ mm}$$

$$M_{\min} = -1666.66 \text{ N} \cdot \text{mm at } x = 1225.00 \text{ mm}$$

Flexural Stresses at $M_{\max}$

$$\text{Distance from NA to top} = 34.84 \text{ mm}$$

$$\text{Distance from NA to bottom} = 41.43 \text{ mm}$$

$$\sigma_{\text{top}} = -5.78 \text{ MPa (compression)}$$

$$\sigma_{\text{bot}} = 6.88 \text{ MPa (tension)}$$

Material Capacities

$$\sigma_{\text{tensile}} = 30 \text{ MPa}$$

$$\sigma_{\text{compressive}} = 6 \text{ MPa}$$

$$\tau_{\max} = 4 \text{ MPa}$$

$$\tau_{\text{glue}} = 2 \text{ MPa}$$

Factor of Safety (Material - Flexural)

$$\text{FOS}_{\text{compression (top)}} = 1.038$$

$$\text{FOS}_{\text{tension (bottom)}} = 4.362$$

Shear Stress at Maximum Shear Location ($x = 1224.87$ mm)

$$\begin{aligned}V_{\max} &= -257.27 \text{ N} \\Q \text{ at centroid} &= 6193.28 \text{ mm}^3 \\b \text{ at centroid} &= 2.54 \text{ mm} \\\tau \text{ at centroid} &= 1.499 \text{ MPa}\end{aligned}$$

Shear Stress at Glue Joints

Glue joint 1 at $y = 75.0$ mm:

$$\begin{aligned}Q &= 4343.90 \text{ mm}^3 \\b &= 112.54 \text{ mm} \\\tau_{\text{glue}} &= 0.024 \text{ MPa}\end{aligned}$$

Factor of Safety (Shear)

$$\begin{aligned}\text{FOS}_{\text{shear (centroid)}} &= 2.668 \\\text{FOS}_{\text{glue (joint 1)}} &= 84.26\end{aligned}$$

Buckling Capacities

$$\begin{aligned}\text{Case 1 (top flange inside, 4 edges): } \sigma_{\text{crit}} &= 3.685 \text{ MPa} \\\text{Case 2 (top flange overhang, 3 edges): } \sigma_{\text{crit}} &= 23.491 \text{ MPa} \\\text{Case 3 (web flexural): } \sigma_{\text{crit}} &= 29.430 \text{ MPa} \\\text{Case 4 (web shear): } \tau_{\text{crit}} &= 5.084 \text{ MPa}\end{aligned}$$

Factor of Safety (Buckling)

$$\begin{aligned}\text{FOS}_{\text{case 1 (top flange inside)}} &= 0.637 \\\text{FOS}_{\text{case 2 (top flange overhang)}} &= 4.062 \\\text{FOS}_{\text{case 3 (web flexural)}} &= 5.089 \\\text{FOS}_{\text{case 4 (web shear)}} &= 3.390\end{aligned}$$

2.2 Plots and Diagrams

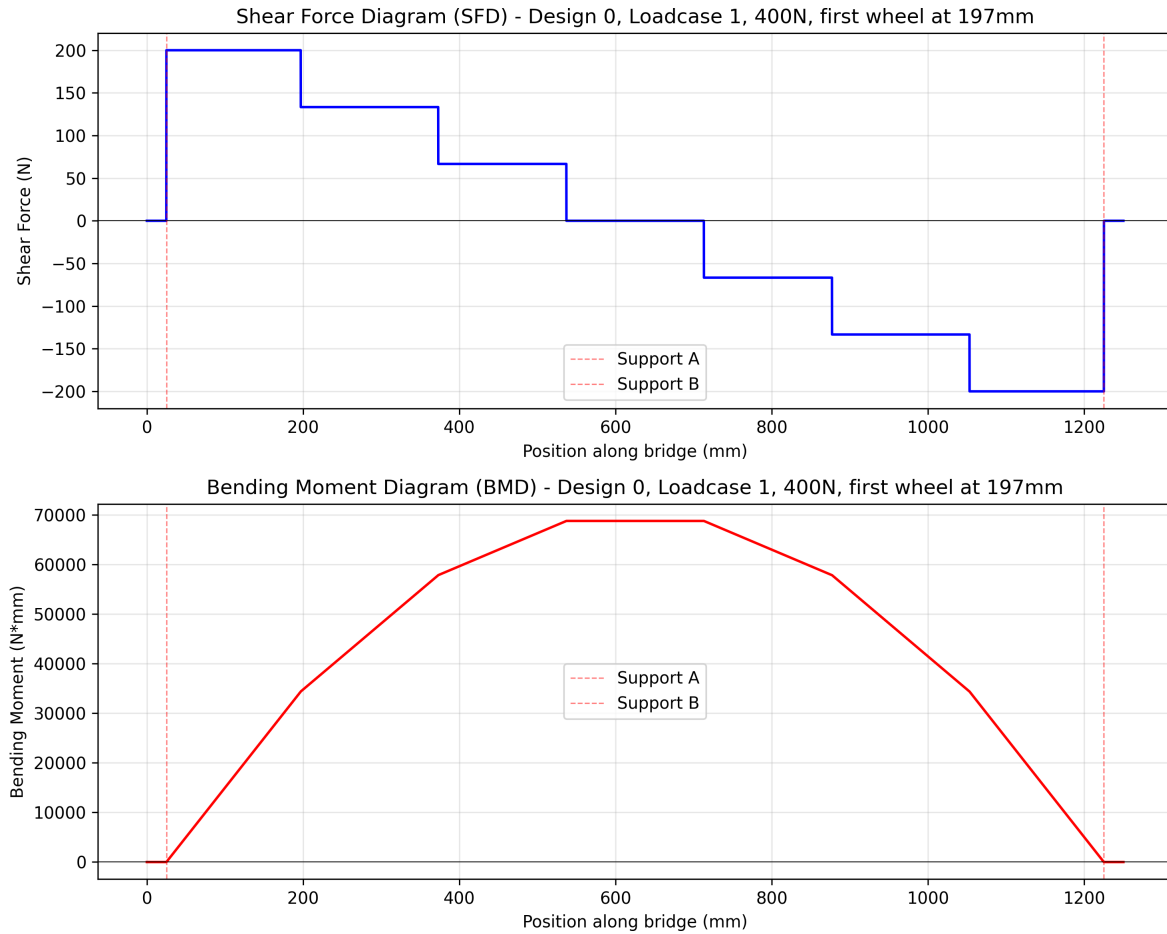


Figure 4: Shear force diagram (SFD) and bending moment diagram (BMD) for Design 0, Load Case 1, train position at 197mm

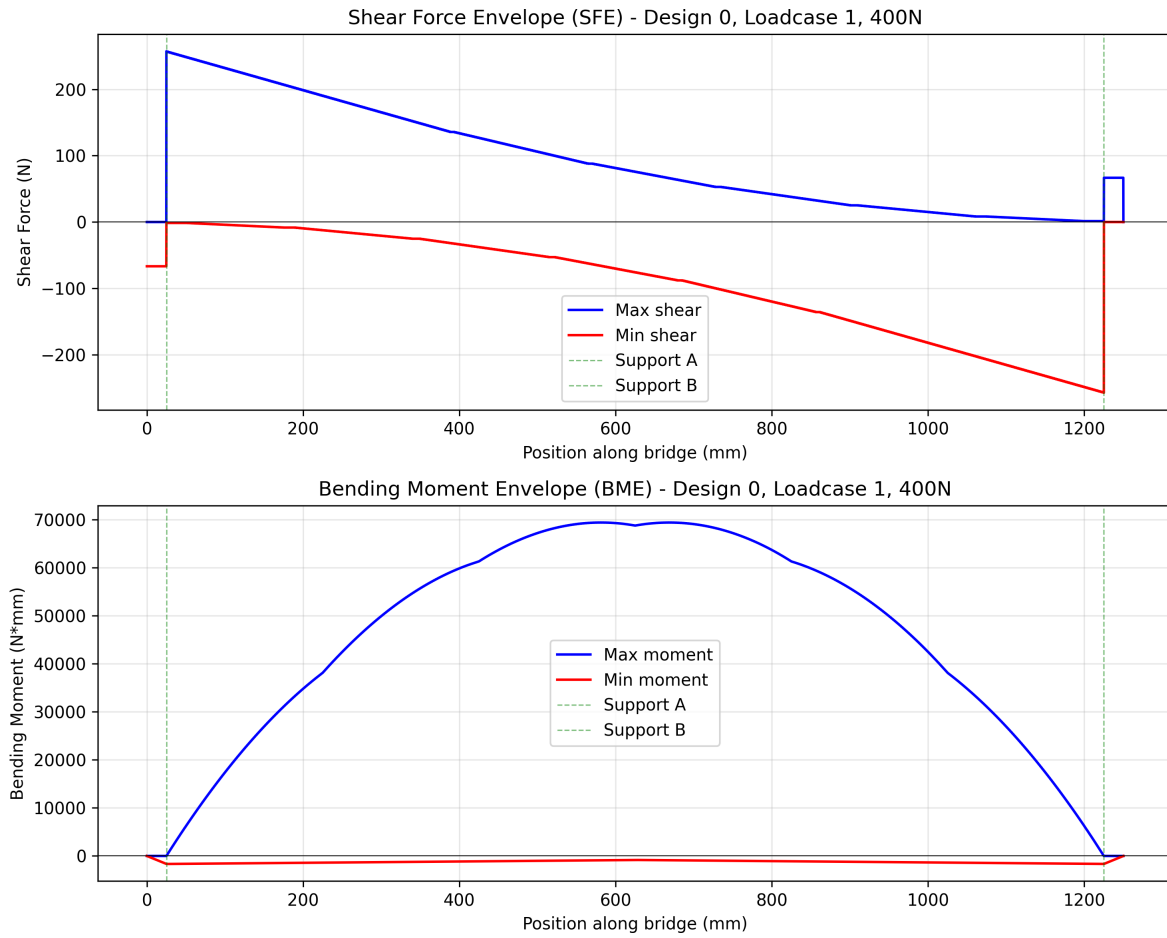


Figure 5: Shear force envelope (SFE) and bending moment envelope (BME) for Design 0, Load Case 1

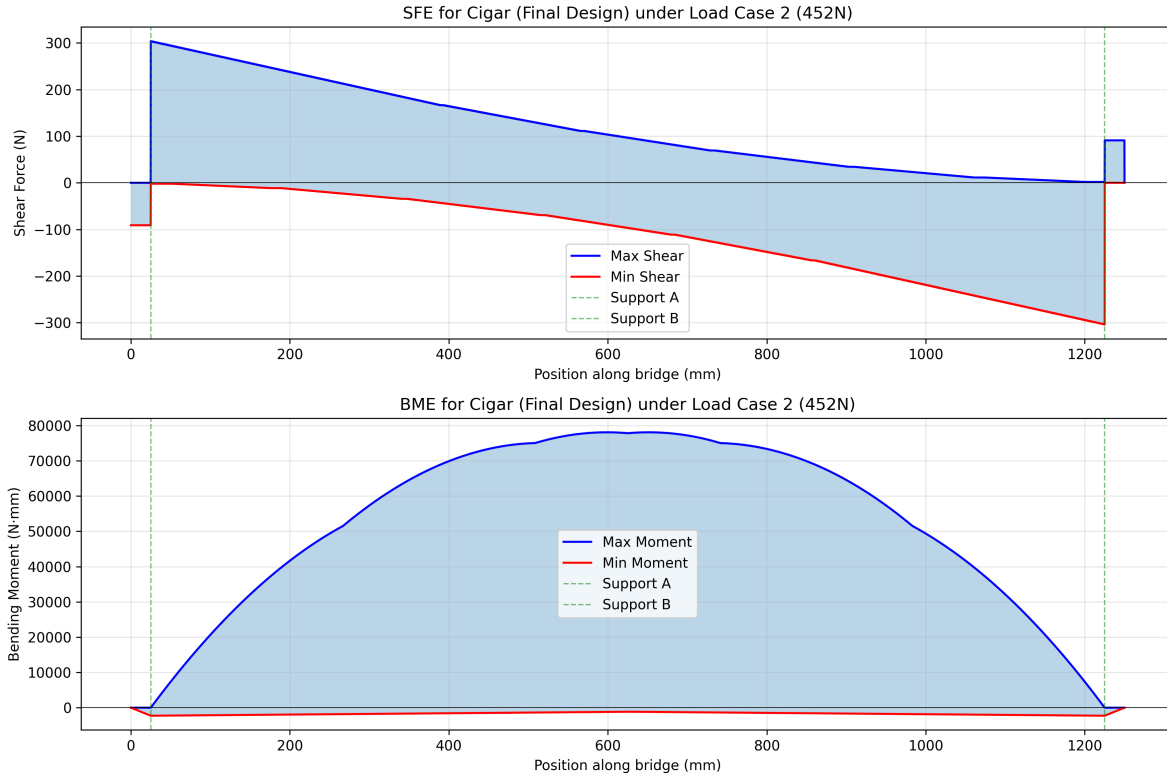


Figure 6: SFE and BME of final design under Load Case 2 (452N base loading)

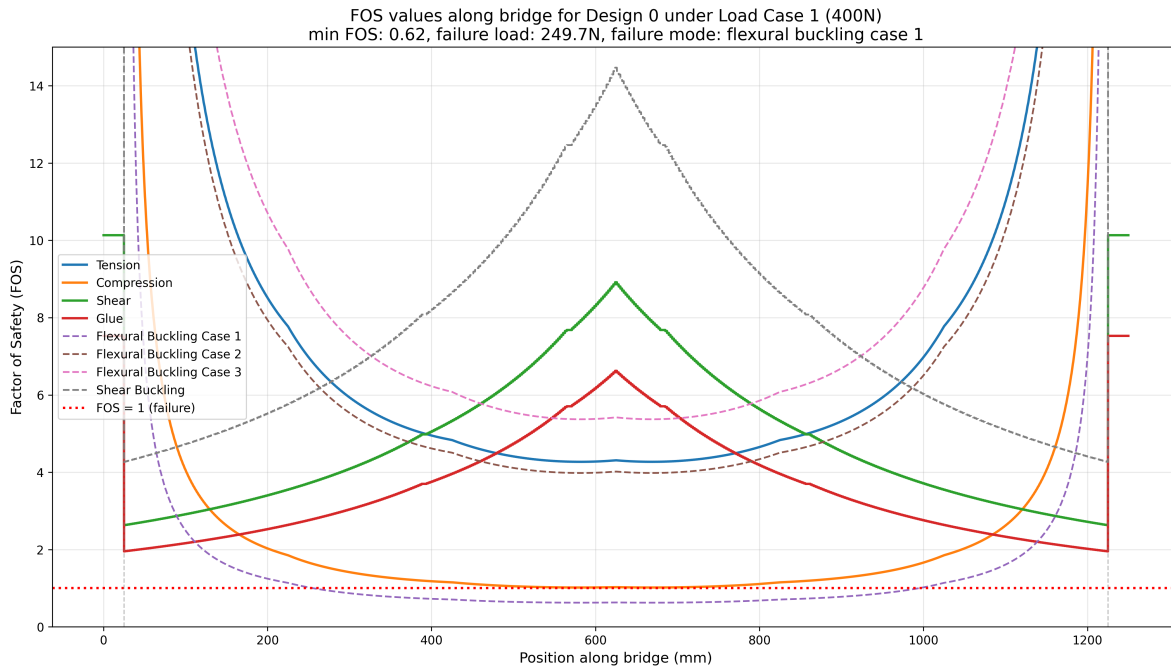


Figure 7: FOS values along the bridge for Design 0, Load Case 1

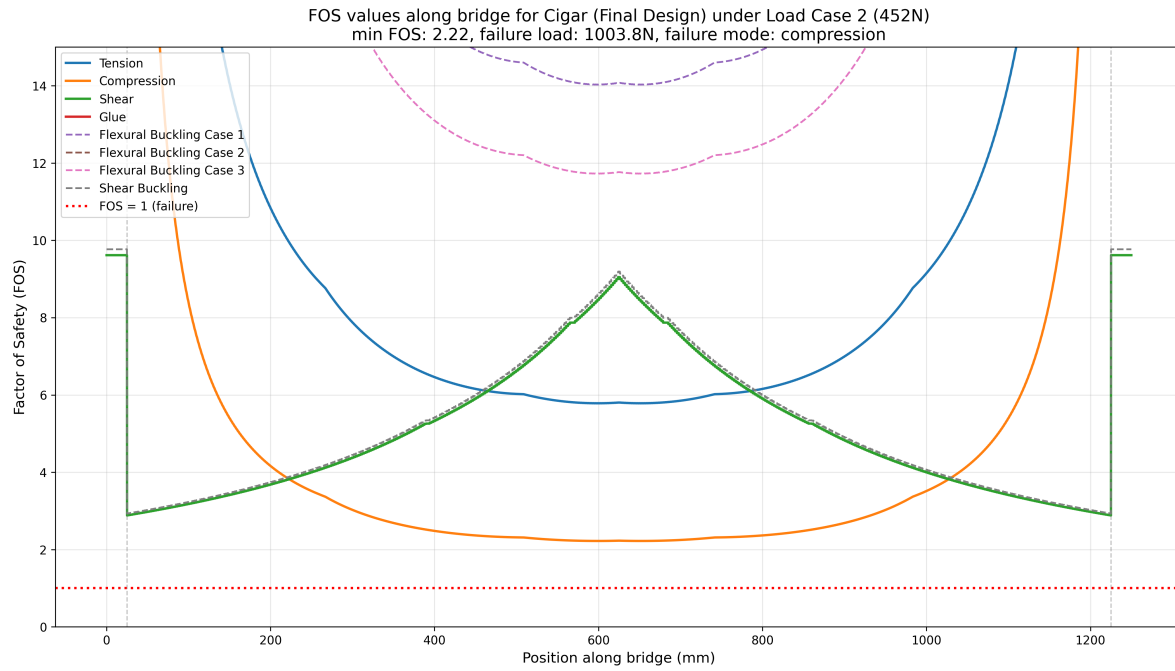


Figure 8: FOS values along the bridge for final design under Load Case 2 (452N)

2.3 Code Repository

All the code: **Repository:** https://github.com/Julian-Moncarz/bridge_project_calc_code